ARE HUMAN-CONSTRAINED LANGUAGE MODELS MORE HUMAN-LIKE? EVALUATING ALGORITHMIC CONVERGENCE ACROSS NEURAL AND BEHAVIORAL MEASURES

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ABSTRACT

Do Large Language Models (LLMs) and biological brains converge onto similar computational mechanisms during language processing? Marr's tri-level framework reveals substantial mismatches between humans and LLMs at the computational level (communicative intent versus next-token prediction) and the implementational level (biological circuits versus artificial architectures); whether the two nevertheless converge at the algorithmic level remains open. To test for it, cognitive scientists have built "human-like" models constrained by human cognitive limitations (e.g., working memory bottlenecks) or by developmentally plausible training exposure (e.g., the BabyLM challenge). Here we ask whether such constraints make a model more human-like. We evaluate 118 frozen models, crossing a memory-limited architecture with a developmental data budget, on 18 behavioral and neural datasets, using hidden-state embeddings and word surprisal as predictors of human responses, and we compare the groups at matched size and perplexity. We find no evidence that either constraint, alone or combined, improves alignment beyond what scale provides. At the last layer, models with both constraints are indistinguishable from standard LLMs, and at the best layer they are clearly worse; a memory limit alone never helps. The one positive coefficient, child-scale training data on embeddings at the last layer, is absent for surprisal and at the best layer. Alignment is predicted by scale for embeddings and by language-modeling quality for surprisal. Predictive power does not show whether representations converge, but it indicates that scale and language-modeling quality matter more than the human-inspired constraints we tested. We make our evaluation framework available as the LLM-Human Convergence Benchmark.

1 INTRODUCTION

Do humans and Large Language Models (LLMs) process language in a similar way? David Marr's tri-level framework provides a principled lens to address this question by decomposing information-processing systems into three distinct levels (Figure 1A): the computational level (what is the goal of the computation?), the algorithmic level (what representations and step-by-step procedures are used?), and the implementational level (how are these steps physically realized?) (Marr, 1982). To evaluate whether LLMs serve as meaningful models of human cognition, we can analyze both systems across these three levels. At the computational level, LLMs are primarily optimized for next-word prediction over massive text corpora. But is next-token prediction truly what the human brain does? While predictive processing plays a central role in human language (Hale, 2001; Levy, 2008; Smith & Levy, 2013; Shain et al., 2024), next-token prediction is at best a rough proxy; biological language processing is driven by broader communicative intent, pragmatic inferencing, multi-step predictive horizons, and complex social or goal-directed rewards (Bender & Koller, 2020; Andreas, 2022). At the implementational level, clear physical and structural distinctions seem to exist. Although artificial neural networks drew early inspiration from neuroscience, modern LLMs have diverged significantly from their biological counterparts. Standard LLMs rely on global self-attention mechanisms across vast context windows (Vaswani et al., 2017), uniform layerwise Transformer blocks that lack the dedicated modularity of the human cortical language network, non-local

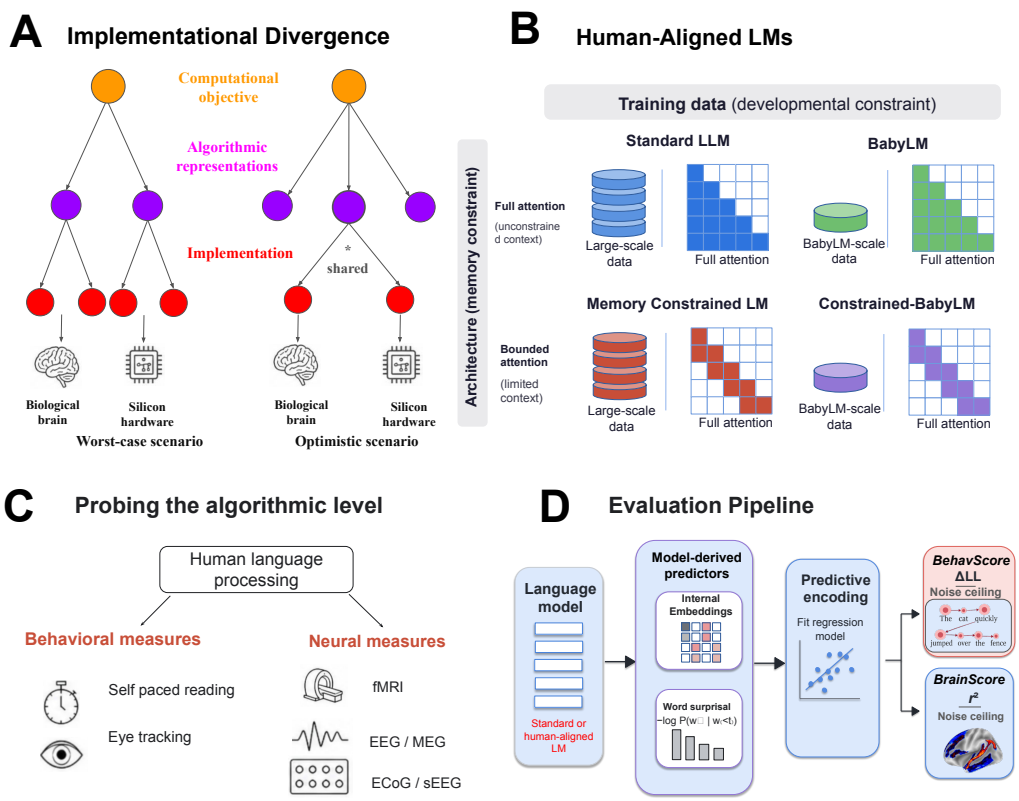


Figure 1: Overview of the study. (A) Marr's tri-level framework applied to humans and LLMs: even where the two systems share the same algorithmic-level representations (optimistic scenario), they may still diverge at the computational and implementational levels. (B) Human-aligned language models introduce either architectural constraints (limited attention, memory decay) or data constraints (developmentally limited training exposure) relative to standard LLMs. (C) Human language processing is probed through behavioral measures (self-paced reading, eye-tracking) and neural measures (fMRI, EEG/MEG, ECoG) at the algorithmic level. (D) LLM-derived predictors – internal representations and word surprisal – are used to predict held-out behavioral and neural data; the resulting scores are called BehavScore and BrainScore.

backpropagation rather than biologically plausible local learning rules (e.g., STDP), and training exposures spanning trillions of tokens – orders of magnitude beyond a human lifetime (Gilkerson et al., 2017; Warstadt et al., 2023a).

Given these differences at both the computational level (Level 1) and implementational level (Level 3), do LLMs and humans converge onto similar representations and processing steps at the algorithmic level (Level 2)? One can map out a spectrum of potential outcomes regarding their algorithmic alignment (Figure 1A). At the pessimistic extreme, these mismatches drive LLMs and human brains to adopt fundamentally different algorithmic steps to solve linguistic tasks. At the optimistic extreme – aligned with the Platonic Representation Hypothesis (Huh et al., 2024) – shared environmental pressures and task demands force biological and artificial systems to converge onto the same underlying representations and algorithms despite their structural differences. Whether LLMs and human brains actually converge at the algorithmic level remains an open empirical question.

Unlike engineering goals focused purely on task performance, cognitive scientists and neuroscientists aim to build models that emulate and explain human-like cognition. To guarantee that artificial models converge on mechanisms closer to human language processing at the algorithmic level, researchers have begun incorporating explicit biological and developmental constraints. On the architectural side, researchers introduce human cognitive limitations, such as working memory capacity

bottlenecks (Lewis & Vasishth, 2005), recurrent processing dynamics (Michaelov et al., 2024a), or localized attention structures (de Varda & Marelli, 2024; Kuribayashi et al., 2022). On the data side, initiatives such as the BabyLM Challenge restrict training corpora to developmentally realistic, child-scale linguistic input (10M to 100M words) rather than web-scale text (Figure 1B; Warstadt et al., 2023a;b). However, whether these human-like constraints actually succeed in driving algorithmic similarity remains an open question. It is entirely unclear if introducing such inductive biases genuinely aligns model mechanisms with human processing, or if standard, unconstrained LLMs already achieve comparable algorithmic convergence simply through task-scale optimization.

How can we empirically determine which algorithms are used by humans and LLMs during language processing? In humans, algorithmic steps are empirically probed through behavioral and neural measures (Figure 1B). Specifically, in psycholinguistics, behavioral measures typically include error rates, error types, self-paced reading times, and eye-tracking fixations (Futrell et al., 2021; Frank et al., 2013; Luke & Christianson, 2018; Siegelman et al., 2022). These offer insights into processing difficulty and moment-by-moment cognitive load. In neuroscience, algorithmic dynamics are captured across complementary spatio-temporal scales: functional Magnetic Resonance Imaging (fMRI) provides whole-brain spatial coverage of, e.g., representational similarity (Schrimpf et al., 2021); Electroencephalography (EEG) tracks precise temporal dynamics, such as the N400 and P600 event-related potentials (Michaelov et al., 2024b; Federmeier et al., 2007); and intracranial electrophysiology (e.g., ECoG) provides joint spatial precision and millisecond-level signal resolution directly from cortical circuits (Schrimpf et al., 2021). Combined, these multi-modal measurements provide the means to evaluate the neural representations and computations underlying language processing. In artificial neural networks, internal mechanisms can be analyzed directly via model interpretability methods; however, given the immense complexity of these black-box representations, interpreting them in isolation remains challenging. Here, rather than relying on internal model interpretability, we test for algorithmic convergence by examining whether LLM-derived metrics, such as token surprisal and internal contextual embeddings, can accurately predict behavioral and neural measurements recorded from humans. Evaluating this predictive power across different model classes allows us to answer the central question of this study: whether humans and LLMs converge on similar algorithmic solutions, and in particular, whether human-constrained models achieve significantly greater convergence than standard LLMs.

We test these questions in a 2×2 design that crosses the architectural and the data constraint (Figure 1B): (1) memory-constrained architectures trained at web scale, (2) the same architectures trained on BabyLM-scale data, (3) standard architectures trained on BabyLM-scale data, and (4) standard, off-the-shelf LLMs with neither constraint. We evaluate 118 frozen models on 18 behavioral and neural datasets that span self-paced reading, eye-tracking, fMRI, EEG and event-related potentials, MEG and intracranial recordings (Figure 1C), using two kinds of predictors, hidden-state embeddings and scalar surprisal (Figure 1D). The four groups differ by orders of magnitude in parameter count, depth and language-modeling quality, so comparing group means would mix the effect of a constraint with the effect of scale. We therefore compare the groups at matched size and perplexity, with a regression on robustly scaled scores, and ask whether a constraint helps beyond what scale and language-modeling quality already provide.

We find no evidence that it does. At matched size and perplexity, models with both constraints are indistinguishable from standard LLMs for embeddings (-0.01 interquartile ranges, SE 0.22) and for surprisal (-0.10, SE 0.10). A memory-limited architecture alone never helps, and when each model's best layer is used it lowers embedding-based alignment. The one positive coefficient, BabyLM-scale training data on embeddings at the last layer, is absent for surprisal and at the best layer. What does predict alignment is scale for embeddings (+0.33 per decade of parameters) and language-modeling quality for surprisal (-0.21 per decade of perplexity). As predictors of human behavioral and neural data, the properties of a language model that matter are therefore its scale and its quality as a language model, not the human-inspired constraints we test.

Our contributions are:

- A common evaluation of 118 language models, from four constraint groups, on 18 behavioral and neural datasets with both embedding and surprisal predictors, which we make available as the LLM-Human Convergence Benchmark.
- A matched-scale analysis that separates the effect of a human-inspired constraint from the effect of model size and perplexity.

- Evidence that neither a memory limit nor a developmental data budget, alone or combined, improves alignment beyond scale, and that scale and perplexity are what predict it.
- Sensitivity analyses showing how the conclusions depend on the layer used to represent a model and on the dependence between models of one family.

2 Related Literature

A growing body of work has asked whether LLM representations line up with human brain activity and behavior during language processing. Model-derived features – most often next-word surprisal or hidden-state embeddings – have been shown to predict fMRI and MEG responses during naturalistic listening and reading (Schrimpf et al., 2021; Caucheteux & King, 2022; Caucheteux et al., 2023), to explain variance in fMRI/ECoG recordings beyond what simpler linguistic features capture (Toneva & Wehbe, 2019), and to track a hierarchy of predictive signals across the cortical language network (Caucheteux et al., 2023). These correlations hold up even for models trained on child-scale corpora (Hosseini et al., 2024) and can be strengthened through instruction tuning (Aw et al., 2024), and recent work has found that recurrent architectures match or exceed Transformers on several of these comprehension metrics despite their very different implementation (Michaelov et al., 2024a). A parallel line of work has probed the internal computations of LLMs directly and reported that intermediate layers already encode next-token distributions in a way that resembles human online processing (Kuribayashi et al., 2025), and that reading times can be decoded from hidden states even when the model was never trained to predict them (Tsipidi et al., 2026). Taken together, these findings motivate the Platonic Representation Hypothesis (Huh et al., 2024): the idea that models trained on different objectives, architectures, and data eventually settle into overlapping representational geometry simply because they are all being pushed to solve related tasks over similar sensory and linguistic statistics.

If shared task pressure is sufficient to produce convergence, why constrain a model at all? One answer, pursued mostly in psycholinguistics, is that human sentence processing operates under resource limits that standard LLMs do not share, and that building those limits in should make a model behave more like a human reader even if it does not change its raw task performance. On the architecture side, this has taken the form of restricting the attention mechanism itself: single-head or otherwise capacity-limited attention (Timkey & Linzen, 2023), distance-based decay that biases attention toward nearby tokens (de Varda & Marelli, 2024; Clark et al., 2025; Thamma & Heilbron, 2025), content-sensitive forget gates (Lin et al., 2025) or stick-breaking attention (Tan et al., 2025), and various memory-bounded or resource-rational formulations of the encoding process (Hahn et al., 2022; Lieder & Griffiths, 2019; Xu et al., 2026; Xu & Futrell, 2026; Rasmussen & Schuler, 2018; Keshev et al., 2025). These architectural choices are usually motivated by classic capacity-limited theories of sentence processing, such as activation-based memory retrieval (Lewis & Vasishth, 2005) and lossy-context surprisal (Futrell et al., 2020). On the data side, the BabyLM Challenge takes the opposite route and leaves the architecture alone but restricts training exposure to the 10-100 million words a child hears by a given age, well below the token budgets of standard LLMs (Warstadt et al., 2023a;b; Hu et al., 2024), building on earlier child-directed-language pretraining work (Huebner et al., 2021). Whether either of these constraints actually buys anything in terms of fit to human data is contested: some studies report that smaller, more human-scale models fit reading times better (Wilcox et al., 2024; Oh & Schuler, 2023a), while others find the opposite, or find that fit to human data and standard language-modeling quality trade off in ways that are not well understood (Oh & Schuler, 2023b; Kuribayashi et al., 2021).

A separate and more technical literature asks how, exactly, one should use an LLM's internal state to predict a given human measurement, largely independent of whether the model itself is human-like. Surprisal remains the dominant predictor of reading times and self-paced reading data (Goodkind & Bicknell, 2018; Wilcox et al., 2020; 2023), with psychometric predictive power now benchmarked systematically across model families and sizes (Kuribayashi et al., 2024). Because surprisal is normally read off a model's final output layer, several groups have instead probed intermediate layers directly – via the logit lens (nostalgebraist, 2020) or the tuned lens (Belrose et al., 2023) – to ask which stage of processing best matches human timing (Tenney et al., 2019); Kuribayashi et al. (2025) report that this intermediate-layer surprisal tracks human reading times increasingly well with depth on several behavioral datasets, well before the final layer. Related work has generalized what counts

as an "anticipation" signal in online processing more broadly (Giulianelli et al., 2024; Opedal et al., 2024).

Closely related and concurrent with our own work, de Varda et al. (2026) compare a low-dimensional effort predictor (surprisal, frequency, and word length) against high-dimensional final-layer embeddings across eight behavioral and five neural datasets, several of which overlap with ours (Natural Stories, Provo, MECO, ZuCo, Tuckute, UCL). They report that behavioral measures are well explained by the low-dimensional effort predictor alone, with embeddings adding little, whereas brain responses require the richer embedding space, and that this split coincides with opposite scaling trends within the GPT-2 family – smaller models fit behavior better, larger models fit the brain better. Their question is largely orthogonal to ours: they hold the model class fixed (a single GPT-2 family) and ask which *predictor* – effort features or embeddings – best explains each modality, whereas we hold the predictor fixed (surprisal or full embeddings, applied identically to every model) and ask which *model class* – architecturally or developmentally constrained versus standard – best explains the data, across a substantially larger sweep of models, model families, and neural/behavioral datasets.

Our study differs from this literature along two axes at once. First, rather than picking a side in the constrained-versus-standard debate, we treat it as an empirical question and put memory-constrained architectures, BabyLM-trained models, and standard off-the-shelf LLMs through the exact same evaluation pipeline, so that any advantage for the "human-like" models has to show up as a direct, apples-to-apples effect rather than being read off separate papers with separate datasets and separate metrics. Second, we do this at a scale that, to our knowledge, has not been attempted for this specific comparison: rather than one behavioral corpus or one neuroimaging dataset, we pool self-paced reading and eye-tracking corpora together with fMRI, EEG/ERP, and intracranial recordings, so that a claim of algorithmic convergence (or its absence) is not an artifact of any single modality, measure, or population.

3 Experimental Design

3.1 Question and design

We ask whether building a human cognitive or developmental constraint into a language model makes its predictions of human behavioral and neural data better than those of a standard, unconstrained LLM. We operationalize "alignment" as predictive power: how well features derived from a frozen model predict held-out human responses. This is a test of convergence in predictions, not of shared mechanism.

Models are organized in a 2×2 design (Section 3.2): an *architectural* factor (a memory-limited architecture or not) crossed with a *data* factor (a developmental, BabyLM-scale training corpus or web-scale text). Coding the two factors as binary variables, MEMORY and DATA, gives three questions: (H1) does a memory limitation change alignment, (H2) does a developmental data budget change it, and (H3) do the two combine additively or interact. Because the four groups differ greatly in parameter count, depth and language-modeling quality (Table 5), a raw comparison of group means would confound the constraint with scale. We therefore compare groups at matched size and perplexity, by regression (Section 3.6).

Every dataset is crossed with every model and with two feature types (Section 3.4), giving a 2×2 of $\{$neural, behavioral$\}$ data $\times$ $\{$embedding, surprisal$\}$ predictors. Everything is identical across models; only per-dataset configurations differ.

3.2 Models

We evaluate 118 frozen models, organized into four groups (Table 5, Appendix B). No model is fine-tuned on any behavioral or neural target.

Standard LLMs (59 models). Off-the-shelf, web-scale autoregressive models with neither constraint: GPT-2, GPT-Neo, OPT, Pythia, OLMo-2, Qwen3, Llama-3.2, Mistral, and the recurrent and state-space models Mamba, Mamba2 and RWKV-4 (11 families, 14M to 66B parameters, 6 to 65 layers), plus the two matched baselines released with the Forgetting Transformer (FoX), which have

ordinary attention and no forget gate (760M parameters). Mamba, Mamba2 and RWKV-4 carry a fixed-size recurrent state but no constraint motivated by human memory, so they are grouped here.

Memory-Constrained (24 models). Architectures that limit how much of the preceding context the model can access or weight, motivated by capacity-limited accounts of human sentence processing such as activation-based retrieval (Lewis & Vasishth, 2005) and lossy-context surprisal (Futrell et al., 2020): distance-penalized attention (ALiBi), forget-gate and stick-breaking replacements for softmax attention, hard n-gram windows, a small bounded-state recurrent network (CBR-RNN), and web-scale models with the same mechanisms (BLOOM, FoX, MoLM). These are trained on web-scale text (15M to 7.1B parameters).

BabyLM (11 models). Standard transformer architectures trained on roughly the number of words a child hears by early adolescence, 10M to 100M words, following the BabyLM Challenge (Warstadt et al., 2023a;b; Hu et al., 2024) (33M to 140M parameters, 11 to 13 layers).

Constrained-BabyLM (24 models). The same memory-limited architectures trained on the BabyLM corpora, i.e. both constraints at once (15M to 48M parameters, 2 to 4 layers).

Keeping the last group separate makes the crossing explicit and lets the regression ask whether the two constraints add up or interact. Pythia and OLMo-2 intermediate checkpoints are available but are not part of the main comparison.

3.3 HUMAN DATA

Every dataset is reduced to the same unit of analysis, a word embedded in its preceding context paired with one or more measured human responses, so any model can be run through any dataset without dataset-specific code. In total we use 18 datasets, which we split into 27 *dataset units*: reading-time measures that come from the same recording (first-pass, go-past and scan-path times) and the three ZuCo reading tasks are counted as separate units, giving 16 behavioral and 11 neural units (Table 6, Appendix C). Units of one corpus share their items and participants and are therefore not independent; we return to this in Section 3.6.

Behavioral datasets. Behavioral measures index processing difficulty indirectly, through the time a reader takes to move past a word.

- **Self-paced reading.** Natural Stories (Futrell et al., 2021), naturalistic narratives deliberately written to contain rare syntactic constructions, and UCL Reading (Frank et al., 2013), 361 sentences read by 117 participants.
- **Eye-tracking.** GECO (Cop et al., 2017), Provo (Luke & Christianson, 2018), the English part of MECO (Siegelman et al., 2022), OneStop (Berzak et al., 2025), and the eye-tracking part of ZuCo (Hollenstein et al., 2018), where participants read isolated sentences under three task instructions (normal reading, sentiment judgment, relation judgment). Depending on the corpus we use first-pass duration, go-past time and scan-path (total dwell) time, each as its own unit.

Neural datasets. Neural measures record language processing directly from the brain at complementary spatial and temporal resolutions.

- **fMRI.** Le Petit Prince (Li et al., 2022) (audiobook listening), Natural Stories fMRI (Shain et al., 2022) (responses pre-averaged within functionally localized language and multiple-demand regions), Tuckute et al.'s language-network dataset (Tuckute et al., 2024) (isolated sentences, including sentences optimized to drive or suppress the network), and Pereira et al. (Pereira et al., 2018) (isolated sentences from two experiments).
- **EEG / ERP.** Broderick et al.'s continuous-listening audiobook corpus (Broderick et al., 2018), and three N400 datasets built around cloze-probability manipulations: a nine-site replication of the classic article/noun expectancy design (Nieuwland et al., 2018), and two studies from the Szewczyk and Federmeier collection (Szewczyk & Federmeier, 2022), Federmeier et al. (2007) (Federmeier et al., 2007) and Hubbard et al. (2019), analyzed separately.

- **Intracranial recordings.** A sentence-reading SEEG dataset (Fedorenko et al., 2016) and a large ECoG dataset recorded during a naturalistic podcast (Zada et al., 2025).
- **MEG.** MEG-MASC (Gwilliams et al., 2022), passive listening to naturalistic stories.

3.4 PREDICTORS: EMBEDDING AND SURPRISAL TRACKS

For each model, dataset and layer ℓ we compute two kinds of predictors from the same forward pass over the full preceding context.

Embedding track. We extract the contextual hidden state of layer ℓ at every word (mean-pooled over its subword tokens) and fit a ridge regression to the human response, with the regularization strength chosen by nested cross-validation and all scores computed on held-out folds (stories, passages, sentences or time chunks, depending on the dataset; Table 7). The score is the held-out correlation between predicted and observed responses. The embedding track uses the full representation, so it measures how much of the signal is linearly decodable from the model. The ridge encoders include no covariates: unlike the surprisal track, they are not given word length or frequency, so lexical information that a model's representation carries can contribute to the score. (Word length and frequency are controlled only in the representational similarity analysis of Appendix H.)

Surprisal track. A scalar surprisal value is obtained at layer ℓ by applying the model's own unembedding matrix to that layer's hidden state (nostalgebraist, 2020). At the last layer this is the model's actual next-token surprisal. We fit a linear mixed-effects model with word length and frequency as baseline predictors and compare it with the model that adds the current and previous word's surprisal. The score is the held-out improvement in log-likelihood per observation, $\Delta LL = (LL^{\text{full}} - LL^{\text{base}})/n_{\text{test}}$, with dataset-specific random-effect structures (Appendix E). A model whose surprisal adds nothing beyond word length and frequency scores 0, regardless of its size.

3.5 LAYER RULE

Models have between 1 and 65 layers, so "the representation of a model" needs a rule that is the same for all of them. Our primary score uses each model's *last* layer, which involves no selection and treats shallow and deep models alike. As a sensitivity analysis, reported in Appendix F, we also use the best layer, defined differently for the two tracks: for embeddings, the layer with the highest held-out score; for surprisal, the layer with the highest training ΔLL, scored on held-out data. Selecting the maximum over a variable number of layers favors deep models, which is why it is not our primary rule.

3.6 SCORES AND STATISTICAL ANALYSIS

BrainScore and BehavScore. We call a model's score on neural datasets its *BrainScore* and its score on behavioral datasets its *BehavScore*; each is defined for both predictor tracks (embedding: held-out correlation; surprisal: held-out ΔLL). These names are used in the Pareto figures of the appendix, where they are the medians over neural and over behavioral dataset units of the score divided by its noise ceiling (Appendix A). The regression below pools all dataset units and uses robustly scaled scores instead.

Scaling. Raw scores (correlations, ΔLL) are not comparable across datasets. For the regression we therefore scale each dataset unit's scores robustly: $s = (x - \text{median})/\text{IQR}$, where the median and the interquartile range are taken over the 118 models of that unit. Coefficients are then expressed in interquartile ranges of model scores within a dataset. Robust scaling needs no estimate of the noise in the data, which is unreliable for several datasets (Appendix A); noise ceilings (van Bree et al., 2025) are used only to describe how much headroom remains, not in the regression.

Matched-scale regression. We regress the scaled score of each (model, dataset unit) pair on the two constraint indicators, the size of the model, and the language-modeling quality of the model on

Table 1: Unadjusted group medians of the robust-scaled score (last layer, all dataset units): units are interquartile ranges of model scores within a dataset, 0 is the median model. These differences mix the effect of a constraint with differences in model size and perplexity (Table 5).

Group	Embedding	Surprisal
Standard LLM	0.11	0.21
Memory-Constrained	−0.10	−0.18
BabyLM	0.20	−0.28
Constrained-BabyLM	−0.52	−0.54

that dataset's text:

$$s_{ij} = \beta_0 + \beta_M \,\text{MEMORY}_i + \beta_D \,\text{DATA}_i + \beta_{MD}\,(\text{MEMORY}_i \cdot \text{DATA}_i) + \beta_S \log_{10} \text{params}_i + \beta_P \log_{10} \text{ppl}_{ij} + \alpha_j + \varepsilon_{ij},$$

where α_j is a dataset-specific intercept, MEMORY $= 1$ for Memory-Constrained and Constrained-BabyLM models, DATA $= 1$ for BabyLM and Constrained-BabyLM models, and Standard LLMs are the reference. β_M and β_D are the effect of one constraint alone, $\beta_M + \beta_D + \beta_{MD}$ is the effect of both (Constrained-BabyLM), and β_S and β_P absorb scale: a constraint helps only if its coefficient is positive *beyond* what size and perplexity predict. We fit this model separately for each predictor track and layer rule, pooling the behavioral and neural dataset units.

Statistical dependence. Models of one family are not independent, and neither is the same model across datasets. We therefore report cluster-robust standard errors, clustered by model family (22 families), alongside the model-based ones. Each constraint group contains only a few families (Appendix B), so the design has limited power to separate a constraint from a family-specific property, and corpus-level dependence between units that share items (for example first-pass and go-past times of one eye-tracking corpus) is not modeled explicitly. We discuss what this means for the conclusions in Section 5.

4 Results

All scores are expressed in interquartile ranges (IQR) of model scores within a dataset (Section 3.6), so a coefficient of 0.30 means 30% of the spread between the 25th- and 75th-percentile model on that dataset. Our primary results use each model's last layer; the best layer is a sensitivity analysis (Section 4.4 and Appendix F). Both predictor tracks are reported throughout, pooling the behavioral and neural datasets. Standard errors are clustered by model family.

4.1 Unadjusted differences between groups

Table 1 gives the median score of each group. With surprisal, Standard LLMs have the highest median (0.21 IQR) and Constrained-BabyLM the lowest (−0.54). With embeddings, BabyLM models have the highest median (0.20, against 0.11 for Standard LLMs) and Constrained-BabyLM again the lowest (−0.52). These raw differences do not tell us what a constraint does, because the groups differ greatly in size, depth and language-modeling quality (Table 5); the next section removes that confound.

4.2 Constraints at matched size and perplexity

Table 2 reports the matched-scale regression. A positive coefficient means that, relative to a Standard LLM of the same size and perplexity, models with that constraint align better with human data.

Both constraints together do not improve alignment. Models with both constraints (Constrained-BabyLM) are not distinguishable from size- and perplexity-matched Standard LLMs, for embeddings (−0.01, SE 0.22) or for surprisal (−0.10, SE 0.10).

Table 2: Matched-scale regression, last layer. Coefficients are shifts in interquartile ranges of model scores within a dataset, relative to a Standard LLM of the same size and perplexity; cluster-robust standard errors (clustered by model family) in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Last row: model–unit rows / dataset units.

Term	Embedding	Surprisal
Memory constraint β_M	0.07 (0.15)	−0.03 (0.08)
Data constraint β_D	0.55 (0.13)***	−0.12 (0.07)
Memory × Data β_{MD}	−0.63 (0.17)***	0.06 (0.08)
Both constraints $\beta_M+\beta_D+\beta_{MD}$	−0.01 (0.22)	−0.10 (0.10)
$\log_{10}$ size β_S	0.33 (0.08)***	0.09 (0.05)
$\log_{10}$ perplexity β_P	0.03 (0.11)	−0.21 (0.04)***
Rows / units	3117/27	3117/27

Memory limits do not help. A memory-limited architecture alone shows no benefit (embeddings +0.07, SE 0.15; surprisal −0.03, SE 0.08). With the best layer, the effect on embeddings turns negative (Section 4.4).

One positive coefficient: BabyLM-scale data on embeddings. The only coefficient that is positive and reliably different from zero is the data constraint on embeddings, +0.55 (SE 0.13). It is cancelled when the two constraints are combined (memory × data −0.63), and it is absent for surprisal (−0.12, SE 0.07) and at the best layer (−0.02, SE 0.08). Unlike the surprisal track, the ridge encoders receive no word length or frequency information (Section 3.4); we return to this in Section 5.

4.3 What predicts alignment: scale and language-modeling quality

The coefficients for size and perplexity in Table 2 show what does predict alignment once the group labels are accounted for, and the two tracks differ.

Embeddings follow scale. Each decade of parameters adds +0.33 IQR (SE 0.08), while perplexity adds nothing reliable (+0.03, SE 0.11).

Surprisal follows language-modeling quality. In the surprisal track, perplexity is the significant predictor: each additional decade of perplexity changes the score by −0.21 IQR (SE 0.04), i.e. lower perplexity means better alignment, whereas size adds little once perplexity is controlled (+0.09, SE 0.05).

4.4 Robustness

Layer rule. Table 8 (Appendix F) repeats the analysis with each model's best layer. The conclusion that no constraint helps holds, and the memory limit becomes harmful for embeddings: the memory coefficient is −0.31 (SE 0.14) and the double constraint −0.49 (SE 0.14). Surprisal results stay close to zero for every constraint coefficient (both constraints +0.02, SE 0.11). Because the best layer is selected as a maximum over layers and deep models have more of them, we regard the last layer as the primary rule and the best layer as the case that is most favorable to deep models.

Statistical dependence. Models of one family are not independent, and each constraint group contains only a few families. Our standard errors are clustered by model family, which is conservative: with plain model-based standard errors many more coefficients would be called significant. The model-based values are in the coefficient table that accompanies the code, but we base our conclusions on the clustered ones. The design cannot separate an architecture effect from a family effect when a family occurs in a single group, which limits what we can claim about individual mechanisms (Section 5).

5 DISCUSSION

We asked whether building a human cognitive or developmental constraint into a language model makes its predictions of human behavioral and neural data better than those of a standard LLM. Across 118 models, 18 datasets and two kinds of predictors, we find no evidence that it does. Once size and perplexity are matched, models with a memory-limited architecture, with a developmental data budget, or with both do not align reliably better with human data than Standard LLMs. What does predict alignment is scale and language-modeling quality, not the human-inspired constraint.

What the results support. Three findings are consistent across analyses. First, a memory limit alone never helps; with the best layer it lowers embedding-based alignment, and the model with both constraints is clearly worse than a Standard LLM. Second, adding the two constraints together never gives a benefit. Third, the variables that do carry signal are scale (for embeddings) and perplexity (for surprisal). In the surprisal track, a better language model gives better surprisal for predicting human data, whereas model size adds little once perplexity is controlled. This pattern is compatible with accounts in which the predictive power of language models for human data follows their task performance (Huh et al., 2024; Shain et al., 2024), but it does not test whether the representations or algorithms converge, which would require evidence beyond predictive power.

The one apparent advantage. The exception is BabyLM-scale training data on embeddings at the last layer, where such models align better with human data than size- and perplexity-matched Standard LLMs (Section 4.2). We do not interpret this as human-likeness. It does not appear for surprisal or at the best layer, and it disappears when the two constraints are combined. A candidate explanation is lexical: the ridge encoders receive no word length or frequency, and the last layer of a small model may retain more word-level information than the last layer of a large model, which is specialized for next-token prediction. We did not test this directly, and the representational similarity analysis (Appendix H), which does control word length and frequency, is not a clean test of it.

The layer rule matters. The size of the memory penalty on embeddings depends on how a model's layer is chosen. With the best layer it is large and the double constraint is clearly worse than a Standard LLM; with the last layer it is small and not distinguishable from zero. Deep models have more layers to choose from, and in our groups depth differs by an order of magnitude (2 to 4 layers for Constrained-BabyLM, up to 65 for Standard LLMs), so the best-layer score is partly a statement about depth. Because depth and size are highly correlated across our models, we could not separate the two, and we therefore draw no conclusion about the size of the memory penalty. What holds under both rules is the absence of any benefit from the constraints.

Limitations. *Power.* Each constraint group contains only a few model families, and the design cannot separate a constraint from a property of the family that implements it. We cluster standard errors by family, which is conservative, and the null results should be read as an absence of evidence for an advantage, not as proof that no advantage exists; a constraint with a small effect would not be detected. *Scope of the constraints.* We test two human-inspired constraints, bounded access to context and a developmental amount of text. Other forms of human-like training, such as curricula, grounded or interactive input, are not covered. *Matched comparison.* The regression matches groups on log size and log perplexity linearly. BabyLM and Constrained-BabyLM models are small and have high perplexity, a region in which few Standard LLMs lie, so part of the comparison relies on extrapolating these terms. *Predictive power is not mechanism.* A model that predicts human data well need not process language like a human, and one that predicts poorly may still share part of the mechanism. *Headroom.* Scores are expressed relative to the spread of the models we evaluate, which does not say how close any model is to the human noise ceiling; for most datasets the best models remain far from it (Appendix A.3). *Data.* Our datasets are largely English and mostly small in participants; the neural datasets rest on a limited number of recordings per modality.

Conclusion. Building memory limits or a child-scale data budget into a language model did not make it a better predictor of human behavioral and neural responses than a standard model of the same size and perplexity. The more promising question is not which constraint to add but which properties of training at scale produce the alignment we observe, and whether those properties can be separated from the constraints that cognitive science has proposed.

Author Contributions

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

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A Robust Scaling and Noise Ceilings

Raw correlations and log-likelihood improvements are not comparable across datasets: a Pearson r of 0.3 might be close to the achievable maximum in a noisy single-trial iEEG recording and far from it in a clean, heavily averaged fMRI dataset. For the regression (Section 3.6) we put every dataset unit on a common scale by robust scaling, $s = (x - \text{median})/\text{IQR}$ over the 118 models of that unit, which needs no estimate of the noise in the data. Noise ceilings, which are the usual remedy, are unreliable for several of our datasets (Tables 3 and 4), so we do not use them in the regression. We use them only to describe *headroom*: how much of the reliable signal in the human data a model recovers. We estimate two kinds of ceiling, one for each predictor track.

A.1 Correlation ceilings: between-subject split-half reliability

For every dataset where a model's score is a correlation between predicted and observed responses (Pearson or Spearman r, at the level of an electrode, voxel, ROI, or channel), we estimate a noise ceiling from the split-half reliability of the human data, following the correction described by van Bree et al. (2025).

Our datasets typically present each item once to each participant, so there is no within-subject repetition to split on. We instead split *across* subjects: for a given dataset and response unit, we repeatedly draw two random, equally sized halves of the subject pool, average each half's response per item, and correlate the two half-averages across items to obtain a split-half correlation r_{sh} – the group-average variant of split-half reliability that van Bree et al. describe for designs without stimulus repetitions. We apply the standard Spearman-Brown correction to extrapolate from a half-sample to the full sample,

$$r_{\text{SB}} = \frac{2\, r_{\text{sh}}}{1 + r_{\text{sh}}},$$

and average r_{SB} over many random subject splits to stabilize the estimate.

It is tempting to treat r_{SB} itself as the ceiling for a model's correlation with the data, but this is the mistake van Bree et al. warn against: r_{SB} is a correlation between two noisy halves, so by the classical correction-for-attenuation logic it bounds *explained variance* (R^2), not the correlation coefficient itself. A model that perfectly recovers the true signal is not itself noisy, so its correlation with the (still noisy) data is attenuated only once rather than twice, giving

$$r_{\max} = \sqrt{r_{\text{SB}}}.$$

We therefore compare every model correlation against $\sqrt{r_{\text{SB}}}$, and every model R^2 against r_{SB} directly, reporting a model's normalized score as the fraction of this ceiling it recovers ($r_{\text{model}} / \sqrt{r_{\text{SB}}}$). Using r_{SB} as a ceiling for a raw correlation, as roughly sixty percent of the papers surveyed by van Bree et al. do, would systematically understate how far a model actually is from ceiling, especially at the low-to-intermediate reliability that characterizes most of our neural datasets. Ceilings are computed separately per response unit, using the same correlation family – Pearson or Spearman – as the corresponding model score, so a score and its ceiling always live on the same scale.

A.2 Delta log-likelihood ceilings: the split-half filler-model design

For the surprisal track, where the quantity of interest is the log-likelihood improvement (ΔLL) a layer's surprisal contributes over a covariate-only baseline, a correlation-based ceiling does not directly apply. We instead adapt the split-half procedure Timkey et al. (2026) use to contextualize their own filler-sentence regression models (their Appendix, Section 3B, "Filler model goodness-of-fit"): we ask how much ΔLL a predictor could achieve in the best possible case, given only the same information available to a single word-level scalar like surprisal – one value per item, no participant- or trial-level access.

We construct this best-case predictor directly from the data. For a given dataset, we randomly split the subject pool into two equal halves, A and B. Within A, we compute the per-item mean response (averaged across subjects and, where relevant, electrodes), substitute it for the surprisal predictor in a mixed-effects model otherwise identical to our real full model, and fit it to the held-out half B. Comparing this model's log-likelihood against a baseline model (identical formula, predictor omitted) fit to the same B data gives one estimate of the achievable ΔLL. We swap the halves – using B's per-item means to predict A – and sum the ΔLL from both directions, repeating over 15 random splits and reporting the mean per-observation ΔLL as the ceiling.

Because the substitute predictor is an empirical group average rather than a noise-free oracle, this ceiling already absorbs the same between-subject averaging noise a real surprisal predictor faces, while staying agnostic to any particular theory of processing difficulty – it is simply the most any single per-item scalar could explain, given the data's own reliability. We report a model's raw ΔLL alongside this ceiling and, where useful, as a percentage of it, exactly as Timkey et al. report their filler models' fit in absolute and relative terms. Two details carry over from our main fitting procedure: ceilings are computed out-of-sample, using the same item/chunk train-test split as the corresponding real analysis, and the ceiling model's random-effects structure mirrors the real model's for that dataset – so the only thing distinguishing the two is the substitution of an empirical per-item mean for the model-derived surprisal.

A.3 Reported ceilings

Tables 3 and 4 report every ceiling computed by the two procedures above, with the fraction of the ceiling reached by the best embedding model (last and best layer) and the share of models whose

surprisal ΔLL exceeds the ceiling. For the embedding track, the best models recover between about 13% and 77% of the ceiling at the last layer, and are furthest from it on the continuous neural recordings (MEG-MASC, Broderick, Podcast). For the surprisal track, four ceilings (Pereira, Federmeier et al. 2007, Hubbard et al. 2019, Broderick; marked ‡) are not usable upper bounds: they rest on few subjects per item or on a very small ΔLL, their relative SD is large, and many models exceed them. This is the reason the regression does not rely on ceilings. A low ceiling indicates that the human measurement itself carries limited reliable signal to recover, and a score built on it should be interpreted with that in mind.

Table 3: Delta log-likelihood noise ceilings for the surprisal track, computed by the split-half item-mean-as-predictor procedure of Appendix A.2, following Timkey et al. (2026). Ceiling is per-observation ΔLL; "splits conv." is the number of converged splits out of those requested. "Above" is the share of the 118 models whose last-layer ΔLL exceeds the ceiling. Rows marked ‡ have a relative SD above 50% or more than 10% of models above the ceiling; their ceiling is not a usable upper bound. Ceilings are used only to describe headroom; the regression uses robust scaling (Appendix A).

Dataset	Ceiling	±SD	Rel. SD	Splits conv.	Above	Note
GECO (FP)	0.00866	0.00036	4%	10/10	0%	small N (14 subj.)
GECO (GP)	0.00416	0.00043	10%	15/15	0%	small N (14 subj.)
MECO (FP)	0.02336	0.00368	16%	15/15	0%	
MECO (GP)	0.00797	0.00186	23%	15/15	0%	
MECO (SP)	0.03786	0.00256	7%	15/15	0%	
Natural Stories (SPR)	0.02381	0.00091	4%	15/15	0%	
UCL Reading (SPR)	0.00236	0.00026	11%	15/15	0%	
OneStop (FP)	0.01286	0.00056	4%	15/15	0%	
OneStop (GP)	0.01429	0.00089	6%	15/15	0%	
OneStop (SP)	0.02359	0.00139	6%	15/15	0%	
Provo (FP)	0.03354	0.00150	4%	15/15	0%	
Provo (GP)	0.03596	0.00160	4%	15/15	0%	
Provo (SP)	0.04573	0.00102	2%	15/15	0%	
ZuCo Reading (NR)	0.02782	0.00225	8%	10/10	0%	small N (12 subj.)
ZuCo Reading (SR)	0.03143	0.00155	5%	15/15	0%	small N (12 subj.)
ZuCo Reading (TSR)	0.01248	0.00239	19%	15/15	0%	small N (12 subj.)
Pereira (fMRI)‡	0.00288	0.00233	81%	15/15	0%	small N ($\leq$10 subj.)
Le Petit Prince (fMRI)	0.01015	0.00088	9%	15/15	0%	
Natural Stories (fMRI)	0.02966	0.00394	13%	15/15	0%	
Tuckute (fMRI)	0.00952	0.00132	14%	10/10	0%	
Podcast (ECoG)	0.00619	–	–	–	0%	9 subj.; weighted over 3 regions, 0–600 ms lags
Fedorenko2016 (ECoG)	0.00832	0.00162	19%	10/10	5%	small N (5 subj.)
Nieuwland (N400)	0.01250	0.00120	10%	30/30	0%	
Federmeier et al. (2007) (N400)‡	0.00261	0.00275	105%	15/15	47%	32 subj.; item-mean oracle
Hubbard et al. (2019) (N400)‡	0.00222	0.00121	55%	15/15	18%	32 subj.; item-mean oracle
Broderick (EEG)‡	0.00010	0.00007	70%	60/60	59%	19 subj.
MEG-MASC (MEG)	0.00073	0.00018	24%	15/15	0%	27 subj.

Table 4: Split-half correlation noise ceilings (embedding track, $\sqrt{r_{\mathrm{SB}}}$, Appendix A.1) for every dataset in the embedding/ridge track, and the fraction of that ceiling reached by the best model, with the last layer (our primary rule) and with the best layer. These ceilings are used only to describe headroom; the regression uses robust scaling (Appendix A). Rows marked † have a ceiling below 0.01 – an absolute reliability so low that a high percent-of-ceiling figure reflects very little recoverable signal in the human data itself, not a strong model fit; we report these for completeness but do not treat them as headline results.

Dataset	Modality	Ceiling ($\sqrt{r_{\mathrm{SB}}}$)	Best model, last layer (%)	Best model, best layer (%)
Hubbard et al. (2019) N400 EEG	EEG (N400)	0.708	77%	85%
ZuCo Reading SR	Behavioral	0.819	75%	81%
ZuCo Reading	Behavioral	0.819	74%	79%
MECO SP	Behavioral	0.930	74%	79%
Federmeier et al. (2007) N400 EEG	EEG (N400)	0.693	71%	81%
MECO	Behavioral	0.899	65%	72%
Provo SP	Behavioral	0.943	61%	69%
Provo	Behavioral	0.933	60%	68%
ZuCo Reading TSR	Behavioral	0.819	60%	65%
GECO	Behavioral	0.627	59%	65%
GECO GP	Behavioral	0.529	56%	62%
OneStop SP	Behavioral	0.888	55%	62%
UCL Reading	Behavioral	0.689	46%	56%
MECO GP	Behavioral	0.684	45%	51%
OneStop	Behavioral	0.829	45%	51%
Nieuwland N400 EEG	EEG (N400)	0.887	45%	63%
Natural Stories SPR	Behavioral	0.914	45%	55%
Le Petit Prince fMRI	fMRI	0.912	45%	49%
Provo GP	Behavioral	0.919	40%	47%
OneStop GP	Behavioral	0.776	35%	38%
Tuckute fMRI	fMRI	0.480	30%	34%
Fedorenko2016 ECoG	ECoG	0.711	29%	40%
Natural Stories fMRI	fMRI	0.628	27%	34%
Pereira fMRI	fMRI	0.507	25%	30%
Podcast iEEG	iEEG	0.398	16%	22%
Broderick EEG	EEG	0.108	15%	23%
MEG-MASC (Gwilliams)	MEG	0.097	13%	20%

B Full List of Evaluated Models

Table 5 lists every model evaluated in this benchmark, with its family, transformer/SSM depth, parameter count, and HuggingFace repository, corresponding to the four model groups introduced in Section 3.2.

The Memory-Constrained and Constrained-BabyLM groups (identical architectures, web-scale vs. BabyLM-scale training data) cover five families: architectures that cap the number of attention heads available for long-range dependencies (Timkey & Linzen, 2023); architectures that bias attention toward nearby tokens through a learned or fixed distance penalty, including ALiBi (Press et al., 2021; de Varda & Marelli, 2024; Clark et al., 2025); architectures that replace softmax attention with a content-sensitive forget gate (Lin et al., 2025) or a stick-breaking process (Tan et al., 2025); hard local n-gram windows that can only ever attend to the two, three, or five preceding tokens regardless of what the input contains (Bu et al., 2026); and CBR-RNN (and its variant CBR-RNN-M), a small recurrent architecture that processes context sequentially rather than through attention at all (Timkey & Linzen, 2023). These architectures and their BabyLM-scale-trained counterparts were obtained from the public HuggingFace checkpoints released alongside Bu et al. (2026). The BabyLM group spans submissions and baselines from the BabyLM shared tasks – including the organizers' own baselines (Charpentier et al., 2025), a masked-language-modeling variant (Edman & Fraser, 2025), an instruction-tuned small model (Capone et al., 2025), a mixture-of-experts architecture (Tapaninaho, 2025), and a model trained with interactive storytelling feedback (Mayer Martins et al., 2025)

– alongside the earlier two BabyLM challenges (Warstadt et al., 2023a;b; Hu et al., 2024), spanning causal Transformer baselines, mixture-of-experts variants, and hybrid causal/masked objectives, all likewise obtained from their public HuggingFace repositories. The Standard LLM group spans twelve off-the-shelf families with no human-inspired constraint on either architecture or training data: GPT-2 (Radford et al., 2019), GPT-Neo (Black et al., 2021), GPT-J (Wang & Komatsuzaki, 2021), GPT-NeoX (Black et al., 2022), OPT (Zhang et al., 2022), Mamba and Mamba-2 (Gu & Dao, 2024), Pythia (Biderman et al., 2023), OLMo-2 (Team OLMo et al., 2025), Qwen3 (Yang et al., 2025), Llama-3.2, and Ministral, from around 100 million to tens of billions of parameters and including both standard Transformer and state-space architectures.

Table 5: All 118 models evaluated in this benchmark, grouped by family within the model groups of Section 3.2 (rules separate each fine-grained family). Memory-Constrained models trained on BabyLM-scale data (marked by `babylm` in the repository name) form the Constrained-BabyLM group; all other models listed under Memory-Constrained are trained on web-scale text. The two "FoX-release baseline" rows are the matched baselines released with FoX, with ordinary attention and no forget gate (the Pro variant keeps the output gate and key/value shift of FoX-Pro); they carry no memory constraint and are therefore counted as Standard LLMs. Parameter and layer counts follow the originating release (or the loaded config for the memory-constrained and BabyLM checkpoints; see `neuroai/models/registry.py`). Training-token/word counts are reported only where documented with confidence by the originating source; a blank cell means this was not reliably available rather than zero.

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Standard LLMs					
GPT-2	openai-community/gpt2	124M	13		(Radford et al., 2019)
GPT-2	openai-community/gpt2-medium	355M	25		(Radford et al., 2019)
GPT-2	openai-community/gpt2-large	774M	37		(Radford et al., 2019)
GPT-2	openai-community/gpt2-xl	1.6B	49		(Radford et al., 2019)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-125m	125M	13		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-1.3B	1.3B	25		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-2.7B	2.7B	33		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-j-6b	6B	29	402B	(Wang & Komatsuzaki, 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neox-20b	20B	45	472B	(Black et al., 2022)
OPT	facebook/opt-125m	125M	13	180B	(Zhang et al., 2022)
OPT	facebook/opt-350m	350M	25	180B	(Zhang et al., 2022)
OPT	facebook/opt-1.3b	1.3B	25	180B	(Zhang et al., 2022)
OPT	facebook/opt-2.7b	2.7B	33	180B	(Zhang et al., 2022)
OPT	facebook/opt-6.7b	6.7B	33	180B	(Zhang et al., 2022)
OPT	facebook/opt-13b	13B	41	180B	(Zhang et al., 2022)
OPT	facebook/opt-30b	30B	49	180B	(Zhang et al., 2022)
OPT	facebook/opt-66b	66B	65	180B	(Zhang et al., 2022)
Mamba	state-spaces/mamba-130m-hf	130M	25	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-370m-hf	370M	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-790m-hf	790M	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-1.4b-hf	1.4B	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-2.8b-hf	2.8B	61	300B	(Gu & Dao, 2024)

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Table 5 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Mamba-2	AntonV/mamba2-130m-hf	130M	25	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-370m-hf	370M	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-780m-hf	780M	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-1.3b-hf	1.3B	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-2.7b-hf	2.7B	61	300B	(Dao & Gu, 2024)
Pythia	EleutherAI/pythia-14m-deduped	14M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-31m-deduped	31M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-70m-deduped	70M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-160m-deduped	160M	12	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-410m-deduped	410M	24	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-1b-deduped	1B	17	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-1.4b-deduped	1.4B	25	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-2.8b-deduped	2.8B	33	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-6.9b-deduped	6.9B	33	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-12b-deduped	12B	37	300B	(Biderman et al., 2023)
OLMo-2	allenai/OLMo-2-0425-1B	1B	17	4.0T	(Team OLMo et al., 2025)
OLMo-2	allenai/OLMo-2-1124-7B	7B	32	3.9T	(Team OLMo et al., 2025)
OLMo-2	allenai/OLMo-2-1124-13B	13B	40	5.0T	(Team OLMo et al., 2025)
Qwen3	Qwen/Qwen3-0.6B-Base	600M	29	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-1.7B-Base	1.7B	29	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-4B-Base	4B	37	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-8B-Base	8B	37	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-14B-Base	14B	41	36T	(Yang et al., 2025)
Llama-3	meta-llama/Llama-3.2-1B	1B	17	9T	
Llama-3	meta-llama/Llama-3.2-3B	3B	29	9T	
Llama-3	meta-llama/Llama-3.1-8B	8B	33	15T	
Ministral	mistralai/Ministral-3-3B-Base-2512	3B	29		
Ministral	mistralai/Ministral-3-8B-Base-2512	8B	37		
Ministral	mistralai/Ministral-3-14B-Base-2512	14B	41		
RWKV-4	RWKV/rwkv-4-169m-pile	169M	13		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-430m-pile	430M	25		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-1b5-pile	1.5B	25		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-3b-pile	3B	33		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-7b-pile	7B	33		(Peng et al., 2023)
RWKV-4	RWKV/rwkv-4-14b-pile	14B	41		(Peng et al., 2023)
FoX-release baseline	zhixuan-lin/transformer-llama-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
FoX-release baseline	zhixuan-lin/transformer-pro-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
Memory-Constrained					
CBR-RNN	cclark/cbr-rnn	17.4M	1		(Timkey & Linzen, 2023)
CBR-RNN	cclark/cbr-rnn-m	17.4M	1		(Timkey & Linzen, 2023)

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Table 5 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Stick-Breaking	Lanni-ni/stickbreaking_pile_2layer	27.8M	2		(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_pile_4layer	48.1M	4		(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_10m_2layer	27.8M	2	∼10M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_10m_4layer	48.1M	4	∼10M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_100m_2layer	27.8M	2	∼100M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_100m_4layer	48.1M	4	∼100M w.	(Tan et al., 2025)
ALiBi	Lanni-ni/alibi_pile_2layer	27.4M	2		(Press et al., 2021)
ALiBi	Lanni-ni/alibi_pile_4layer	45.7M	4		(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_10m_2layer	27.4M	2	∼10M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_10m_4layer	45.7M	4	∼10M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_100m_2layer	27.4M	2	∼100M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_100m_4layer	45.7M	4	∼100M w.	(Press et al., 2021)
Forgetting-Transf.	Lanni-ni/forgetting_pile_2layer	27.4M	2		(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_pile_4layer	45.7M	4		(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_10m_2layer	27.4M	2	∼10M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_10m_4layer	45.7M	4	∼10M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_100m_2layer	27.4M	2	∼100M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_100m_4layer	45.7M	4	∼100M w.	(Lin et al., 2025)
Hard 2-gram	Lanni-ni/hard_2gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_100m_2layer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_100m_4layer	28.8M	4	∼100M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_100m_2layer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_100m_4layer	28.8M	4	∼100M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_pile_4layer	28.8M	4		(Bu et al., 2026)

continued on next page

Table 5 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Hard 5-gram	Lanni-ni/hard_5gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_100m_2layer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_100m_4layer	28.8M	4	∼100M w.	(Bu et al., 2026)
BLOOM	bigscience/bloom-560m	560M	25		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-1b1	1.1B	25		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-1b7	1.7B	25		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-3b	3B	31		(BigScience Workshop, 2022)
BLOOM	bigscience/bloom-7b1	7.1B	31		(BigScience Workshop, 2022)
FoX (Forgetting Transf.)	zhixuan-lin/fox-llama-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
FoX (Forgetting Transf.)	zhixuan-lin/fox-pro-760m-longcrawl64-48b	760M	25	48B	(Lin et al., 2025)
MoLM	ibm/MoLM-350M-4B	350M	25		(Shen et al., 2023)
MoLM	ibm/MoLM-700M-4B	700M	25		(Shen et al., 2023)
MoLM	ibm/MoLM-700M-8B	700M	49		(Shen et al., 2023)
BabyLMs					
BabyLM	colinglab/CLASS_IT-140M	140M	13		(Capone et al., 2025)
BabyLM	llm-slice/blm-gpt2s-90M-s42_901M-s42_submission	90M	13		
BabyLM	Jtapsa/moep	80M	13		(Tapaninaho, 2025)
BabyLM	BabyLM-community/BabyLM-2026-Baseline-GPT2-Strict	98.4M	13	∼100M w.	(Charpentier et al., 2025)
BabyLM	BabyLM-community/BabyLM-2026-Baseline-GPT2-Strict-Small	98.4M	13	∼10M w.	(Charpentier et al., 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-causal-focus	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-mixed	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-masked-focus	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-causal-focus	33M	13	∼10M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-mixed	33M	13	∼10M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-masked-focus	33M	13	∼10M w.	(Edman & Fraser, 2025)

C Full List of Evaluated Datasets

Table 6 lists every dataset evaluated in this benchmark, grouped by modality, corresponding to the behavioral and neural categories introduced in Section 3.3.

Table 6: All datasets evaluated in this benchmark, grouped by modality. N items reports the unit each dataset's items are counted in (sentences, passages, screens, stories, electrodes, ...), since this varies by design. Eye-tracking measures: FP = first-pass/gaze duration, GP = go-past time, SP = (single-fixation or total) dwell time. N subj. reflects each study's usable sample after its own quality-control exclusions.

Dataset	Modality	Subj.	Items	Measure(s)	Ref.
Self-Paced Reading					
Natural Stories SPR	SPR	180	10 stories (10,256 word pos.)	raw RT	(Futrell et al., 2021)
UCL Reading	SPR	117	361 sentences	raw RT	(Frank et al., 2013)
Eye-Tracking					
GECO	Eye-tracking	14	588 screens	FP, GP	(Cop et al., 2017)
MECO (English)	Eye-tracking	46	12 passages	SP, FP, GP	(Siegelman et al., 2022)
Provo	Eye-tracking	84	55 texts	SP, FP, GP	(Luke & Christianson, 2018)
OneStop	Eye-tracking	360	330 passages	SP, FP, GP	(Berzak et al., 2025)
ZuCo Reading	Eye-tracking	12	SR/NR/TSR: 400/300/407 sent.	TRT	(Hollenstein et al., 2018)
N400 / ERP					
Nieuwland	EEG (ERP)	334	9-lab RSVP replication	N400	(Nieuwland et al., 2018)
Federmeier et al. (2007)	EEG (ERP)	32	564 items	N400	(Federmeier et al., 2007; Szewczyk & Federmeier, 2022)
Hubbard et al. (2019)	EEG (ERP)	32	192 items	N400	(Szewczyk & Federmeier, 2022)
Continuous EEG					
Broderick	EEG	19	20 segs. ($\sim$196s each)	EEG (1–8Hz)	(Broderick et al., 2018)
MEG					
Gwilliams (MEG-MASC)	MEG	27	4 stories $\times$ 2 sess.	continuous MEG	(Gwilliams et al., 2022)
fMRI					
Le Petit Prince	fMRI	49	9 sections (audio)	BOLD	(Li et al., 2022)
Natural Stories fMRI	fMRI	78	8 narratives (26 fROIs)	BOLD	(Shain et al., 2022)
Tuckute	fMRI	10	2,000 sentences (6 ROIs)	BOLD	(Tuckute et al., 2024)
Pereira	fMRI	$\leq$10	627 sentences (12 ROIs)	BOLD	(Pereira et al., 2018)
iEEG / ECoG					
Podcast	ECoG	9	1 story, >5,000 w. (1,330 electr.)	high-gamma	(Zada et al., 2025)
Fedorenko2016	iEEG	5	52 sentences (416 words)	iEEG (scalar)	(Fedorenko et al., 2016)

D IMPLEMENTATION DETAILS

Data provenance. All datasets were obtained from their public OpenNeuro or OSF repositories under the terms specified therein.

Feature extraction. All language models are loaded in bfloat16 and evaluated as frozen feature extractors, with no gradient updates. For each word, hidden states are mean-pooled over its constituent subword tokens at every layer; unless otherwise noted (Section 3.4), a word's representation is computed with its full preceding sentence or discourse context in the forward pass, rather than in isolation. Surprisal at every layer is obtained via a logit-lens projection, applying each model's own unembedding matrix to the corresponding intermediate hidden state.

Ridge regression. For every dataset, nested cross-validation selects the ridge penalty from a log-spaced grid via an inner search, with predictors standardized by a robust (median/IQR) scaler fit on the training fold only, and outer folds grouped so that no sentence, item, story, or run straddles the train/test boundary. The grid and fold count are tuned per dataset to its sample size and predictor dimensionality rather than fixed globally: Table 7 lists the exact configuration for every dataset. We report mean out-of-fold Pearson and Spearman correlation across the outer folds, and sweep this procedure across every layer of every model to obtain a layerwise profile.

Table 7: Ridge regression configuration for the ridge encoder of each dataset (embedding track): log-spaced α grid (number of grid points in parentheses), outer cross-validation fold count, and the grouping unit used so that no fold splits correlated observations. GroupKFold groups on the listed unit unless marked LOSO (leave-one-unit-out, i.e. fold count equals unit count). Datasets with no embedding/ridge track (OneStop, Szewczyk) are omitted.

Dataset	α grid (points)	Folds	Grouping unit
GECO (FP/GP)	10^{1}–10^{10} (10)	5	screen
MECO (FP/GP/SP)	$10^{0.5}$–10^{10} (12)	5	passage
Provo (FP/GP/SP)	10^{0}–10^{10} (11)	5	text
UCL Reading	$10^{1.5}$–10^{9} (9)	5	sentence
OneStop (FP/GP/SP)	10^{0}–10^{10} (11)	5	passage
Natural Stories SPR	10^{-1}–10^{5} (7)	10 (LOSO)	story
Natural Stories fMRI	10^{-2}–10^{4} (6)	8 (LOSO)	story
ZuCo Reading (NR)	$10^{2.5}$–10^{9} (8)	5	sentence
ZuCo Reading (SR/TSR)	10^{2}–10^{6} (8)	5	sentence
Nieuwland	$10^{3.5}$–10^{10} (6)	10	sentence item
Federmeier et al. (2007), Hubbard et al. (2019)	$10^{1.5}$–$10^{5.25}$ (7)	10	sentence frame
Tuckute	10^{2}–10^{7} (5)	4	sentence
Broderick	$10^{3.5}$–10^{8} (5)	3	audiobook segment
MEG-MASC	$10^{3.5}$–10^{8} (5)	4 (LOSO)	story
Le Petit Prince	10^{3}–$10^{5.3}$ (8)	9 (LOSO)	section
Podcast	$10^{3.5}$–10^{8} (5)	3	contiguous time chunk
Fedorenko2016	$10^{3.5}$–10^{8} (5)	5	sentence
Pereira	10^{1}–10^{6} (8)	5	passage

Temporal response function (TRF) datasets. For the three datasets with a temporal response rather than a single scalar per word – Broderick, Podcast, and Fedorenko2016 – the ridge encoder fits a lag-matrix TRF: one multi-output ridge regression predicts every time lag jointly from the same set of predictors, rather than one regression per lag. Broderick's EEG is downsampled to 32 Hz and epoched from -0.2 to 1.0 s relative to word onset (38 lags); Podcast's ECoG uses the

same 32 Hz rate over a wider -1.0 to 1.0 s window (128 lags). Fedorenko2016 has no temporal axis: its intracranial recordings are already reduced to a single scalar response per (word, electrode) pair, so no lag-matrix step is needed there.

Podcast's per-region mixed-effects model. In addition to its per-electrode ridge encoder, Podcast is also fit with a linear mixed-effects model on the response averaged over the electrodes of each language region (Appendix E), with a random intercept per subject and no electrode term. One model is fit *at each lag* of the 0–600 ms post-onset window (19 lags of the -0.2 to 0.8 s, 32-lag grid), for the final layer of each language model, and the benchmark score is the mean over lags rather than the value at a peak lag, which avoids selecting the lag on the held-out data. The matching noise ceiling is computed lag by lag and region by region with the split-half procedure of Appendix A.2 and averaged over lags in the same way; each region's score is normalized by its own ceiling before the regions are combined (weighted by held-out observations). Held-out partitions are formed from contiguous 30-word chunks of the continuous transcript, the direct word-index analogue of the chunk-based splits used for the fMRI datasets below.

fMRI encoding pipelines. For Le Petit Prince and Natural Stories fMRI, word-level predictors are convolved with a canonical double-gamma HRF (Glover, 1999), implemented via `nilearn`, before ridge or mixed-effects fitting; both use $\mathrm{TR} = 2$ s and discard the first and last several volumes of each run for signal equilibration. Le Petit Prince additionally restricts voxels to the top 25% by split-half reliability before averaging into ten bilateral parcels of the Fedorenko language-network atlas, and both its ridge and mixed-effects analyses are computed on the group-average BOLD signal rather than per subject, following the most-reliable-voxel selection and subject-averaging procedure of Bonnasse-Gahot & Pallier (2024). Natural Stories fMRI aggregates BOLD into 26 functional regions of interest (6 left-hemisphere language-network parcels, 20 bilateral multiple-demand parcels) per Shain et al. (2022). Tuckute's sentence-level design uses only the 1,000-sentence naturalistic condition for encoding and mixed-effects analyses; the two adversarially optimized conditions designed to drive or suppress the language network are evaluated separately, since only 2–3 of the 10 participants saw each of those sentences.

Linear mixed-effects models. All LME models are fit in R via `lme4`. Where out-of-sample evaluation is used, held-out partitions are formed from contiguous chunks of the data (words, sentences, or short time windows depending on the dataset), shuffled with a fixed seed before assignment to train/held-out splits, so that both partitions sample the full range of each recording or corpus rather than, e.g., only its first half. If a model with the full random-effects structure specified in Appendix E fails to converge, we follow the simplification procedure of Barr et al. (2013): the correlation between random slopes is dropped first, and if convergence still fails, the slope itself is dropped, leaving a random-intercept-only term. This fallback is applied identically across every layer of a given model so that log-likelihood comparisons remain valid.

E Linear Mixed-Effects Formulas

This appendix specifies the linear mixed-effects (LME) models underlying the surprisal track analyses of Section 3.4, in standard `lme4` formula notation. Each model is fit once per transformer layer: a *baseline* model containing only nuisance covariates, and a *full* model that additionally includes the layer's surprisal, with the two compared via the log-likelihood difference described in Appendix A.2. All continuous predictors are z-scored within dataset; `surprisal_z` denotes the per-layer logit-lens surprisal of the evaluated model, and `prev_surprisal_z` the same quantity for the preceding word (a spillover control).

E.1 Self-paced reading and eye-tracking

Reading times and fixation durations across Natural Stories, UCL Reading, GECO, MECO, Provo, OneStop, and ZuCo Reading are modeled with a common specification, with word length, frequency, and sentence position as nuisance covariates and by-subject random slopes:

```
baseline: fdur ~ sentpos_z + wlen_z + freq_z +
                 (1 + sentpos_z + wlen_z | subject)
```

```
full: fdur ~ sentpos_z + wlen_z + freq_z + surprisal_z
+ prev_surprisal_z +
             (1 + sentpos_z + wlen_z + surprisal_z +
prev_surprisal_z | subject)
```

E.2 N400 EVENT-RELATED POTENTIALS

Nieuwland is modeled following the specification of its originating study rather than our shared self-paced-reading/eye-tracking structure, since its multi-site design calls for a different random-effects treatment. Following Nieuwland et al. (2018), laboratory is entered as a fixed rather than random effect, since a random-effect variance is not reliably estimable from only nine sites; there is no electrode term, as N400 amplitude is already averaged across centro-parietal channels in this dataset, nor a frequency covariate, matching the original analysis:

```
baseline: n400 ~ lab + (1 | subject) + (1 | item)
full: n400 ~ lab + surprisal_z + (1 | subject) + (1 |
item)
```

Federmeier et al. (2007) and Hubbard et al. (2019) are fit separately, one model per study, with the specification of the analysis scripts of the Szewczyk and Federmeier collection (Szewczyk & Federmeier, 2022). N400 amplitude is pre-averaged over four centro-parietal electrode clusters into one value per trial, so there is no electrode term. The nuisance covariates are the pre-stimulus baseline amplitude (`bline_z`), log frequency, word position in the sentence, orthographic neighbourhood (`old20_z`) and concreteness, and random slopes are uncorrelated (`||`); `item` is the sentence frame:

```
baseline: n400 ~ bline_z + freq_z + pos_start_z +
old20_z + concr_z +
                 (1 + bline_z + pos_start_z || subject)
+ (1 + bline_z || item)
full: n400 ~ bline_z + freq_z + pos_start_z + old20_z +
concr_z + surprisal_z +
             (1 + bline_z + pos_start_z || subject) +
(1 + bline_z || item)
```

E.3 CONTINUOUS EEG AND MEG

For Broderick, one model is fit per layer on the data of all subjects pooled, with a random intercept for subject. `region` denotes four anatomical scalp quadrants (left, right, posterior, occipital), the same for every subject.

```
baseline: eeg ~ wlen_z + freq_z + (1 | subject) + (1 |
region)
full: eeg ~ wlen_z + freq_z + surprisal_z +
prev_surprisal_z + (1 | subject) + (1 + surprisal_z
| region)
```

For MEG-MASC, the response is the mean amplitude in a 250–450 ms post-onset window, averaged within six sensor groups (`region`, the same grouping for every subject); one pooled model is fit per layer over all subjects, with a random intercept for subject. The nuisance covariates are word duration, word length, frequency and the duration of the previous word:

```
baseline: y ~ worddur_z + wlen_z + freq_z +
prev_worddur_z + (1 | subject) + (1 | region)
full: y ~ worddur_z + wlen_z + freq_z + prev_worddur_z +
surprisal_z + prev_surprisal_z +
          (1 | subject) + (1 + surprisal_z | region)
```

E.4 fMRI

For Natural Stories fMRI, `story` is a crossed random intercept (there are only eight stories) and the single random slope is on `subject`; no spillover term is included, since HRF convolution already blends the contributions of adjacent words into a single BOLD sample. Dropping the frequency covariate from both models gives the variant without frequency.

```
baseline: bold ~ wlen_z + freq_z + sentpos_z + (1 |
subject) + (1 | story)
full: bold ~ wlen_z + freq_z + sentpos_z + surprisal_z
+
             (1 + surprisal_z | subject) + (1 | story)
```

For Le Petit Prince, one model is fit per layer on the BOLD signal of all subjects; `region` denotes one of ten bilateral language-network parcels from the Fedorenko functional atlas, `section` one of the nine narrative chapters, and `subject` enters as a random intercept.

```
baseline: bold ~ wlen_z + freq_z + sentpos_z + (1 |
region) + (1 | section) + (1 | subject)
full: bold ~ wlen_z + freq_z + sentpos_z + surprisal_z
+
             (1 + surprisal_z | region) + (1 |
section) + (1 | subject)
```

For Tuckute, the response is a single BOLD scalar per (subject, ROI, sentence), so word length and frequency are averaged over the sentence (`mean_wlen_z`, `mean_freq_z`) and surprisal is computed over the whole sentence rather than word-by-word. The model is fit on the naturalistic sentence condition only.

```
baseline: y ~ mean_wlen_z + mean_freq_z +
              (1 + mean_wlen_z + mean_freq_z | subject)
+ (1 | roi) + (1 | sentence_id)
full: y ~ mean_wlen_z + mean_freq_z + surprisal_z +
          (1 + mean_wlen_z + mean_freq_z + surprisal_z |
subject) + (1 | roi) +
          (1 + surprisal_z | sentence_id)
```

For Pereira, which uses the same per-sentence design as Tuckute but with sentences of variable length, we follow the specification of Lin & Schuler (2026) directly: sentence length (`sentlen_z`) is retained as an additional nuisance covariate alongside mean word length and frequency, `roi` is a 12-level crossed random effect (the language-atlas parcellation used across all subjects), and surprisal is again computed over the whole sentence rather than word-by-word.

```
baseline: y ~ sentlen_z + mean_wlen_z + mean_freq_z +
              (1 + sentlen_z + mean_wlen_z + mean_freq_z
| subject) + (1 | roi) + (1 | sentence_id)
full: y ~ sentlen_z + mean_wlen_z + mean_freq_z +
surprisal_z +
          (1 + sentlen_z + mean_wlen_z + mean_freq_z +
surprisal_z | subject) + (1 | roi) +
          (1 + surprisal_z | sentence_id)
```

E.5 Intracranial recordings

For Fedorenko2016, electrodes are patient-specific, so `electrode` is nested within `subject` rather than treated as a crossed effect. Podcast is instead fit per region on region-averaged electrodes, so no electrode term is needed: for each language region (anterior temporal, inferior frontal, posterior temporal; left-hemisphere electrodes assigned to the Fedorenko language parcels, with the orbital part of the inferior frontal gyrus merged into the inferior frontal region and the middle frontal gyrus dropped because it has fewer than four subjects), the high-gamma response is averaged over

Table 8: As Table 2 but with each model's best layer (embedding: highest held-out score; surprisal: highest training ΔLL).

Term	Embedding	Surprisal
Memory constraint β_M	−0.31 (0.14)*	0.05 (0.08)
Data constraint β_D	−0.02 (0.08)	−0.05 (0.08)
Memory × Data β_{MD}	−0.16 (0.13)	0.02 (0.07)
Both constraints $\beta_M+\beta_D+\beta_{MD}$	−0.49 (0.14)***	0.02 (0.11)
$\log_{10}$ size β_S	0.29 (0.06)***	0.17 (0.05)***
$\log_{10}$ perplexity β_P	−0.15 (0.08)	−0.17 (0.03)***
Rows / units	3117/27	3117/27

the region's electrodes within each subject and word, and the model is fit separately at each lag of the 0–600 ms post-onset window (19 lags on the 31.25 ms grid), at the last layer only; no spillover term is included. Responses are standardized with statistics computed on the training chunks only (per electrode, then per subject and region after averaging) and applied unchanged to the held-out chunks, so no information from the held-out data enters the scaling. The headline score is the mean over lags of the held-out ΔLL per observation of each region, divided by that region's own noise ceiling and then averaged across regions weighted by their number of held-out observations.

```
baseline: y ~ wlen_z + freq_z + (1 | subject)
full: y ~ wlen_z + freq_z + surprisal_z + (1 | subject)
```

For Fedorenko2016, spillover is included, along with a richer by-subject random-slope structure:

```
baseline: y ~ wlen_z + freq_z + (1 + wlen_z + freq_z |
subject) +
              (1 | subject:electrode)
full: y ~ wlen_z + freq_z + surprisal_z +
prev_surprisal_z +
          (1 + wlen_z + freq_z + surprisal_z +
prev_surprisal_z | subject) +
          (1 + surprisal_z + prev_surprisal_z |
subject:electrode)
```

F Best-Layer Regression

Table 8 repeats the matched-scale regression of Table 2 with each model's best layer instead of its last layer. For embeddings, the best layer is the one with the highest held-out score; for surprisal, the layer with the highest training ΔLL, scored on held-out data. Models differ greatly in depth (1 to 65 layers) and the best-layer score is a maximum over a different number of layers per model, which favors deep models.

G Pareto Fronts for Other Layer Selections and Feature Modes

Figure 3 reports the neural-behavioral trade-off of Figure 2 for the remaining combinations of feature mode and layer selection: surprisal features at the best layer (the layer with the largest training-set ΔLL), and embedding features at the last and at the best layer. Surprisal scores are normalized by the ΔLL ceilings and embedding scores by the correlation ceilings (Appendix A); the axes, point sizes and colors follow Figure 2.

H Representational Similarity Analysis: Methods

The ridge encoding models of the embedding track (Section 3.4) have a number of free parameters that grows with the hidden size d of the model, so a score that increases with model size may

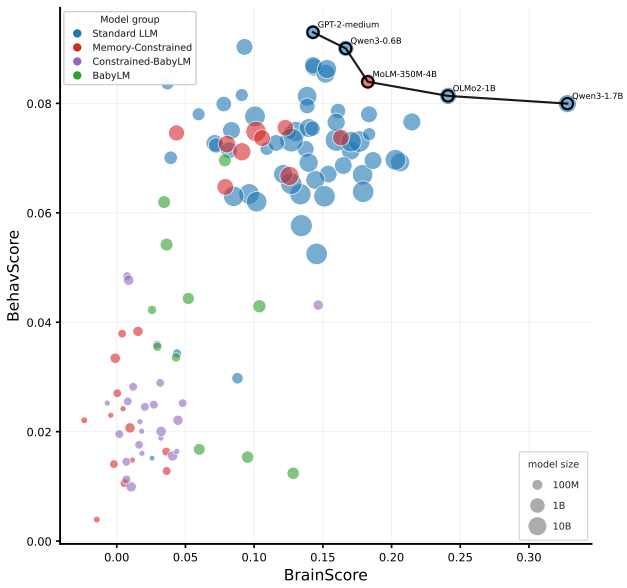


Figure 2: Neural-behavioral trade-off (BrainScore vs. BehavScore, last layer, surprisal features, both axes divided by the ΔLL noise ceiling) across all evaluated models. Point size is model parameter count; color is model group (Appendix B); the black-outlined path traces the Pareto-optimal frontier.

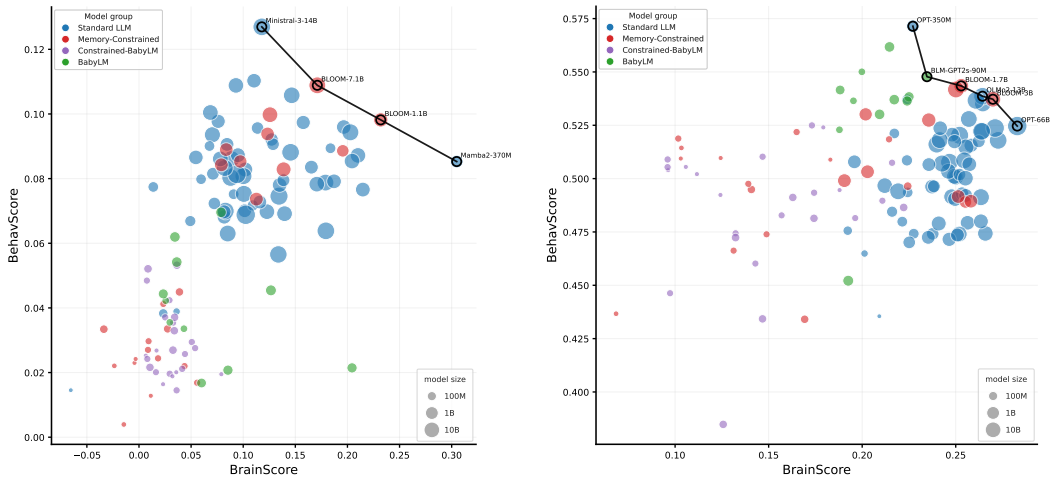
reflect the capacity of the regression rather than a better representation. Representational similarity analysis (RSA; Kriegeskorte et al., 2008) removes this dependence: whatever d is, a model yields a stimulus-by-stimulus dissimilarity matrix (RDM), and the downstream regression contains a single model predictor, with no per-model weights and no tuned hyperparameter. We therefore run RSA as a complementary analysis to both tracks (hidden-state embeddings and scalar surprisal), on the same 118 models, stimuli and datasets as the main benchmark. The following subsections describe the dissimilarity matrices (H.1), the scores computed from them (H.2), cross-validation (H.3), the noise ceilings (H.4) and the aggregation (H.5). All RSA results use the last layer of each model.

H.1 Dissimilarity matrices

Let i, j index the stimuli of a dataset (words, sentences or items, depending on the dataset; see Table 6). An RDM is symmetric with a zero diagonal, so each pair of stimuli is used once, which gives $n(n-1)/2$ pairs for n stimuli. All pairs are used for every dataset except GECO and OneStop, whose word counts (43,296 words for GECO, 937 million pairs) make the full matrix too large to hold per model; for these two we draw a random subset of 30 million pairs once and use this same subset for every model, both tracks and the noise ceilings. With all pairs, the sums needed by the regression are computed as products of the dense upper-triangular distance matrix with the item-fold indicator matrices, which gives exactly the same statistics as the pair-list implementation. In both cases all regressions and correlations operate on the pairs of the same set for the model, the confounds and the ceiling.

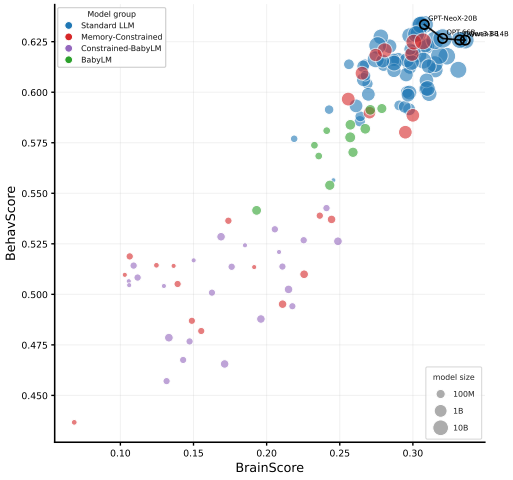
Embedding RDM. For stimulus i, let $h_i \in \mathbb{R}^d$ be the last-layer hidden state of the model, extracted as in the main benchmark (word-level, or the pooled sentence representation for sentence datasets). Raw last-layer states of transformers contain a handful of "massive-activation" dimensions with a large mean and variance, which dominate cosine and correlation distances (Timkey & van Schijndel, 2021). Following Timkey & van Schijndel (2021), we therefore z-score every feature across the stimuli scored in the dataset,

$$\tilde{h}_{ik} = \frac{h_{ik} - \mu_k}{\sigma_k}, \qquad \mu_k = \frac{1}{n}\sum_i h_{ik}, \quad \sigma_k^2 = \frac{1}{n}\sum_i (h_{ik} - \mu_k)^2,$$



(A) Surprisal, best layer.

(B) Embedding, last layer.



(C) Embedding, best layer.

Figure 3: Neural-behavioral trade-off (BrainScore vs. BehavScore) for the layer selections and feature modes not shown in Figure 2. Point size is model parameter count; color is model group (Appendix B); the black-outlined path traces the Pareto-optimal frontier.

(constant features are set to zero), and define

$$D^E_{ij} = 1 - \mathrm{corr}(\tilde{h}_i, \tilde{h}_j),$$

the correlation distance between the standardized vectors. This standardization is applied identically to every embedding RSA, behavioral and neural. In Timkey & van Schijndel (2021), μ_k and σ_k are computed over a corpus of token representations; here the corpus is the set of stimuli scored in each dataset. Because the statistics are estimated on the same stimuli that are compared, the transformation is transductive (test stimuli contribute to μ_k, σ_k), although no response data are involved.

Surprisal RDM. For the surprisal track, $s_i = -\log p(w_i \mid w_{<i})$ is the surprisal of the (critical) word, or the summed surprisal of the sentence for sentence-level datasets, exactly as used by the mixed-effects analyses, and

$$D^S_{ij} = |s_i - s_j|.$$

Behavioral response RDM. For each behavioral measure (first-pass, go-past and total reading times, self-paced reading times, etc.), t_i is the mean of the measure for item i over all participants

who read it (grand average), and

$$D^B_{ij} = |t_i - t_j|.$$

Neural response RDM. Every response unit (an ROI, channel or time lag, depending on the dataset) is treated like a separate dependent variable of the surprisal mixed-effects model: its response y_i to stimulus i gives

$$D^N_{ij} = |y_i - y_j|.$$

Standardizing the responses of a unit before taking differences only multiplies this matrix by a constant, and neither the RSA correlation nor the held-out ΔLL changes under such a rescaling.

Nuisance RDMs. The surprisal mixed-effects models include word length and word frequency as covariates; we mirror them with $|\Delta\,\text{length}|$ and $|\Delta\,\text{Zipf frequency}|$ between the two stimuli of a pair (for sentence datasets: sentence length, mean word length and mean word frequency). For Federmeier et al. (2007) and Hubbard et al. (2019) we additionally include $|\Delta\,\text{position}|$, $|\Delta\,\text{OLD20}|$ and $|\Delta\,\text{concreteness}|$, to match the covariates of their published mixed-effects formulas. Random-effect structures have no counterpart on a subject-averaged item target and are not reproduced.

H.2 Scores

Two scores are computed from the same RDMs.

(a) RSA correlation. The classical RSA score of model m is the Pearson correlation between the upper-triangular entries of its RDM and of the response RDM,

$$R_m = \operatorname{corr}\big(v(D^m),\, v(D^{\text{response}})\big),$$

where $v(\cdot)$ denotes the upper-triangular entries (all or the sampled pairs).

(b) RDM regression and held-out ΔLL. For each dataset, track, model and unit we fit two ordinary-least-squares models on the pair entries,

$$\text{base:}\quad D^{\text{response}}_{ij} = \beta_0 + \sum_k \beta_k\, C^{(k)}_{ij} + \varepsilon_{ij},$$

$$\text{full:}\quad D^{\text{response}}_{ij} = \beta_0 + \sum_k \beta_k\, C^{(k)}_{ij} + \gamma\, D^m_{ij} + \varepsilon_{ij},$$

where $C^{(k)}$ are the nuisance RDMs and $D^m \in \{D^E, D^S\}$ is the model RDM (all entries z-scored before fitting), with i.i.d. Gaussian errors whose variance is the maximum-likelihood estimate on the training pairs. The score is the gain in held-out log-likelihood relative to the confound-only model, per test pair,

$$\Delta LL_m = \frac{LL^{\text{test}}_{\text{full}} - LL^{\text{test}}_{\text{base}}}{n_{\text{test pairs}}}.$$

This mirrors the headline of the surprisal track ($\Delta\ell\ell_{\text{test}}/n_{\text{test}}$ against a covariate-only baseline), so that the two analyses are on a comparable "gain beyond confounds, out of sample" scale. Since the full model has exactly one model predictor regardless of the model's width, the regression capacity does not depend on the number of embedding dimensions.

H.3 Item-wise cross-validation

Entries of an RDM are not independent: each stimulus enters $n-1$ pairs. Splitting pairs at random would leak, since a test pair shares a stimulus with many training pairs. We therefore split *stimuli*, following Schütt et al. (2023) and Jozwik et al. (2017): items are randomly assigned to K folds, and for each fold f the model is fit on the pairs whose two items are both outside f and evaluated on the pairs whose two items are both inside f; pairs with one item in f are discarded. The number of folds depends on the number of items n as in Schütt et al. (2023): $K = 2$ for $n < 12$, 3 for $12 \le n < 24$, 4 for $24 \le n < 40$ and 5 for $n \ge 40$. The test log-likelihoods are summed over folds, and the procedure is repeated with new random partitions (10 repeats for the neural datasets, 30 for the behavioral ones); the reported ΔLL is the mean over repeats. The partitions are fixed per dataset

and unit, so that all models and both tracks are scored on identical train/test pairs, and base and full models are always scored on the same test pairs.

Pairs sharing an item remain correlated inside the training and test sets, which violates the independence assumption of the Gaussian likelihood. This inflates absolute ΔLL values by roughly the same amount for all models; we therefore use ΔLL to rank models and do not interpret its absolute scale.

H.4 Noise ceilings for RSA

As for the ridge and mixed-effects tracks (Section A), RSA scores are expressed relative to the reliability of the human data. Because the quantity scored here is a gain in held-out log-likelihood on a pair-wise distance matrix, the ceiling is computed in the same RDM space, with the same item folds, the same nuisance RDMs and the same Gaussian pair likelihood as the models. All ceilings use 15 random splits.

Split-half ΔLL ceiling. We adapt the split-half variance ceiling of Timkey et al. (2026) (Section A.2) to RDM space. The participants are randomly split into two equal halves A and B (15 splits). Half A's item-mean RDM $|m_i^A - m_j^A|$ is used as the predictor of half B's item-mean RDM $|m_i^B - m_j^B|$ in the full model of Section H.2, in place of a model RDM, and compared with the same confound-only baseline on the same test pairs; the roles of A and B are then swapped, the two total ΔLL are summed and divided by the total number of test pairs, and the ceiling is the mean over splits (with 5 fold partitions per split). Items must be observed in both halves. For neural datasets the split is over subjects and is carried out separately for each unit (ROI, channel, or ROI $\times$ lag), so each unit has its own ceiling; for the N400 datasets there is a single unit. The predictor is built from only half of the participants, so the ceiling is conservative, and no Spearman–Brown correction is available on the ΔLL scale.

RSA-correlation ceiling. For the RSA correlation R_m we use the split-half reliability of the response RDM, in the same spirit as the correlation ceilings of Section A.1: r_{half} is the Pearson correlation between the entries of the two half-sample RDMs (averaged over the 15 splits), extrapolated to the full sample by the Spearman–Brown formula, $r_{\text{SB}} = 2r_{\text{half}}/(1 + r_{\text{half}})$, and the ceiling for a correlation is $\sqrt{r_{\text{SB}}}$ (van Bree et al., 2025).

Normalized score. The normalized RSA score of model m on a dataset is the median over its units u of the RSA correlation divided by the ceiling of that unit,

$$S_m = \underset{u}{\text{median}} \ \frac{R_{m,u}}{\rho_u}, \qquad \rho_u = \sqrt{r_{\text{SB},u}}.$$

Units whose split-half reliability r_{half} is not positive have no defined ceiling and are excluded from the median. The ΔLL is normalized in the same way, by the split-half ΔLL ceiling C_u of the unit. Because the ceilings are conservative, ratios above 1 can occur.

H.5 Aggregation

Scores are computed per dataset, and across datasets we report the median of the per-dataset scores, as in the rest of the benchmark.

H.6 Pareto fronts of the RSA scores

Figure 4 shows, for each of the 118 models, the normalized RSA score on neural data (BrainScore) against the normalized RSA score on behavioral data (BehavScore), separately for the surprisal and the embedding track. BrainScore is the median over the eight neural datasets of the dataset score S_m defined above (Pereira and Tuckute restricted to their language ROIs, Podcast restricted to the language parcels, Broderick, MEG-MASC, and the three N400 datasets). BehavScore is the median over the 16 behavioral measures of R_m/ρ, with ρ the RSA-correlation ceiling of each measure. A model is on the Pareto front if no other model has both a higher BrainScore and a higher BehavScore.

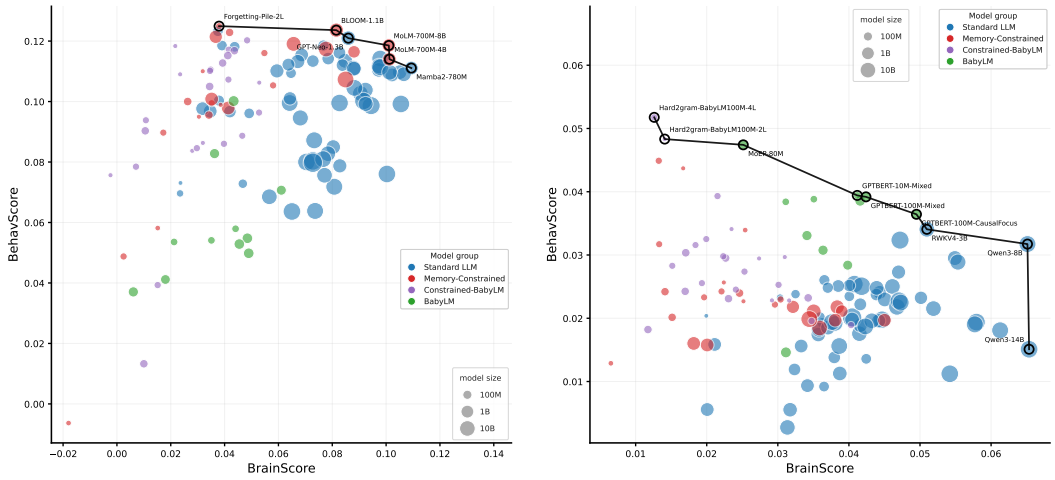


Figure 4: Pareto fronts of the RSA scores, last layer. Each point is a model (color: model group; area: number of parameters). BrainScore and BehavScore are medians over datasets or measures of the RSA correlation divided by the noise ceiling of each unit. Left: surprisal track ($|\Delta s|$ model RDM). Right: embedding track ($1-$correlation of standardized hidden states). Circled points and the black line show the Pareto front.

Only units with a positive ceiling enter the medians. Table 9 gives the number of units of each neural dataset that have a positive ceiling for the RSA correlation and for the ΔLL (the latter requires that the half-sample predictor improves on the confound-only baseline).

Table 9: Number of response units with a positive noise ceiling, out of all units of the dataset.

Dataset	Units	RSA ceiling > 0	ΔLL ceiling > 0
Pereira (language ROIs)	24	20	9
Tuckute (language ROIs)	13	13	9
Podcast (language parcels $\times$ lags)	128	122	100
Broderick (sensor banks)	4	4	2
MEG-MASC (regions)	6	6	4
N400 datasets (each)	1	1	1

Figure 5 shows the same analysis for the held-out ΔLL of the RDM regression. The scores are normalized per unit by the split-half ΔLL ceiling C_u, and units with a non-positive ceiling are excluded from the median. For the behavioral measures the score is the full-sample last-layer ΔLL divided by the ceiling of the measure.

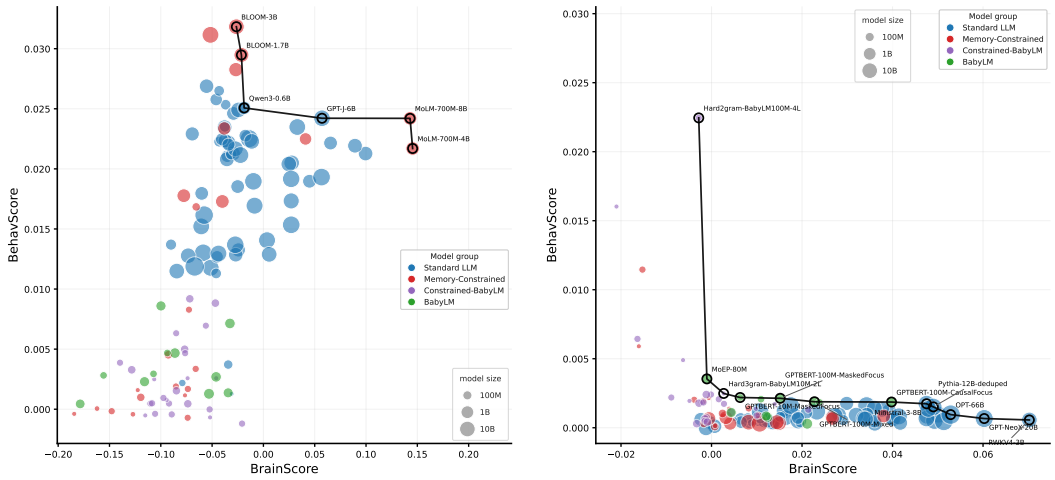


Figure 5: Pareto fronts of the held-out ΔLL of the RDM regression, last layer. BrainScore and BehavScore are medians over datasets or measures of the ΔLL divided by the split-half ΔLL ceiling of each unit. Left: surprisal track. Right: embedding track. Conventions as in Figure 4.